

ECON485 Fall 2025
Midterm Exam 21.11.2025 – KEY 2 (AI)

Student Number		Grading	
Name and Surname		Questions 1	 out of 15
Student Signature	THIS KEY IS PREPARED TO DEMONSTRATE HOW STRAIGHTFORWARD AI USE GENERATES CORRECT BUT NOT NECESSARILY PERFECT DESIGNS.	Question 2	 out of 25
		Question 3	 out of 15
Notes: 1. Exam duration is 150 minutes, please take your time. Some questions and parts are easy to answer once you do the previous ones. In particular, you must repeatedly reuse earlier designs (ERD, normalized tables) when answering Q3 and Q4. 2. All questions are based on the Case Study which includes non-trivial business rules. Please read it carefully. Take notes, draw diagrams or small tables. Use the back of pages if necessary for your notes. These notes will not be graded, so you can make mistakes there. You are expected to make mistakes. 3. Some questions test your skills as an analyst. As analysis is an iterative process, the questions are designed to make you add more and more to your DB design. The remaining questions test your skills in the technical aspects (ie. SQL statements, and discussion of concepts). 4. You are allowed to use your notes and check up on the course Github for previous examples. 5. You are allowed and encouraged to use AI tools we covered in the course. 6. Please write your name on the corner of each page. Also if you need more space, feel free to use the back of pages.		Question 4	 out of 35
		Question 5	 out of 10
		Question 6 (Bonus)	 out of 10
		Question 7 (Bonus)	 out of 10
		Total	 out of .100

CASE STUDY

The Catholic Church served as the primary custodian of vital records throughout medieval Europe, especially from the early Middle Ages until the late fifteenth century, because secular governments were often too weak, fragmented, or illiterate to maintain systematic documentation.

Literacy resided almost entirely with the clergy, and parish priests together with monastic scribes therefore recorded baptisms, marriages, and deaths. In the earlier centuries these entries appeared as informal notes written in the margins of missals, on loose sheets, or inside monastic cartularies that mainly copied charters and privileges.

From about the thirteenth century onwards, many parishes began keeping more organised ledgers, although practices varied widely by region. Only after the Council of Trent in 1563 did the Church mandate standardised, separate registers for baptisms (including godparents' names), marriages, and burials, with duplicate copies sent to the bishop. Most entries were in Latin and remained extremely

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brief – a typical baptismal record might read no more than “John, son of William the smith, baptised on the feast of St Michael, 1350”.

Communication between local churches and Rome followed a strict hierarchy. Everyday matters – ordinations, institutions to benefices, or simple marriage dispensations – were handled by the diocesan bishop and recorded in his register. Only cases involving high-ranking nobles, politically sensitive unions, or impediments in the third or fourth degree of kinship were referred upward, sometimes as far as the papal curia.

Requests travelled by trusted messenger or professional courier, and papal replies arrived as bulls (formal documents with a lead seal), briefs, or ordinary letters. The journey from England or Germany to Rome (or to Avignon during the fourteenth-century papal residence there) could easily take six months to a year, so wealthy families often kept permanent agents at the curia to speed the process.

Marriage required careful scrutiny because the Church prohibited unions within certain degrees of consanguinity (blood relationship) and affinity (relationship by marriage). Before the Fourth Lateran Council of 1215 the prohibition extended to the seventh degree, making almost every noble marriage technically incestuous and in need of dispensation. Innocent III reduced the forbidden degrees to the fourth (first cousins), a rule that has remained in force ever since. Direct-line relationships and sibling marriages were prohibited by divine law and could never be dispensed.

For ordinary parishioners the priest announced banns on three successive Sundays so that anyone aware of an impediment – kinship, pre-contract, or vows of chastity – could come forward.

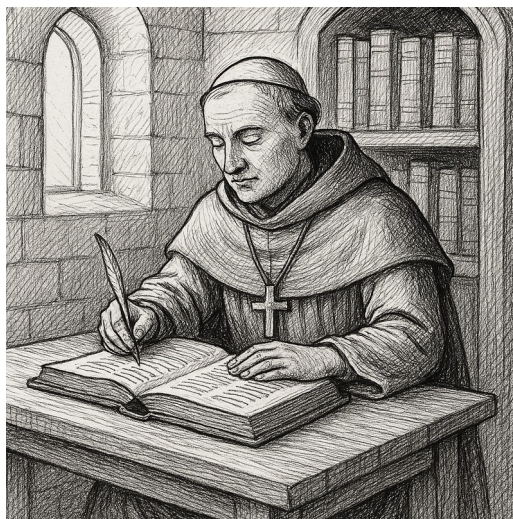


Figure 1: Medieval priest working on marriage records

Noble and royal couples, however, frequently needed dispensations for political alliances, and they submitted detailed petitions (often accompanied by a family tree) to the bishop or directly to Rome, explaining why the union served the peace of Christendom or prevented greater evils. Once granted,

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the dispensation was recorded and the marriage could proceed; without it the union was invalid and any children illegitimate, with devastating consequences for succession.

Beyond baptisms, marriages, and deaths, the medieval Church maintained many other records that affected secular life. Bishops' registers documented the appointment of priests, visitations to parishes, and the proving of wills, and the granting of marriage licences.

Monasteries kept cartularies copying every land grant and privilege, together with necrologies listing the death dates of benefactors so that prayers could be said in perpetuity. Ecclesiastical courts heard cases of defamation, breach of marriage contract, and bastardy, generating voluminous acts that often supply the only evidence of family relationships when parish registers are lost.

Until 1858 in England (and comparable dates elsewhere) the Church retained jurisdiction over probate, so wills and detailed inventories of possessions passed through church courts. Priests frequently acted as clerks for manorial courts, recording land transfers, fines, and inheritance among peasants. Some parishes, especially after Trent, compiled status animarum or "soul books" – household-by-household censuses noting every inhabitant's age and sacramental status – that function as early local population surveys. Tithe lists and taxation records drawn up for church levies provide yet more quasi-censuses.

Taken together, these ecclesiastical documents formed the nearest thing medieval Europe possessed to a centralised population database. A person's baptismal entry proved legitimacy and noble status; the parents' marriage record (or the absence of a valid one) determined whether heirs could inherit land, titles, or thrones; death dates fixed the moment when property changed hands or a new reign began.

Kings, dukes, and barons relied on church archives to defend their claims against rivals, and disputed or forged entries could ignite civil wars. The Church's monopoly on literacy and record-keeping therefore placed it at the very heart of medieval political stability: its registers were simultaneously spiritual, legal, and dynastic instruments upon which the fate of families, estates, and entire kingdoms frequently depended.

-- This space is intended for your note taking --

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Question 1 (Analysis, 15 points)

There is an empty table below about the types of records kept by the church and their purposes. Fill out the table for the the most important five types of records kept by the Church. Hint: There are more than 5, relate importance to the following questions by checking up what's ahead.

WHEN USED STRAIGHTFORWARD, AI FOCUSES ON THE INPUT TEXT ONLY AND PRIORITIES COME FROM HOW THE CONCEPTS ARE PRESENTED IN THE TEXT. FOR THIS QUESTION, THIS APPROACH PRODUCES SUFFICIENT RESULTS.

Record Type	What it contains	Purposes & use cases
Baptismal Records	Individual's name, baptism date, parents' names (including father's profession), and godparents' names.	Proved legitimacy and noble status . Served as early local population surveys.
Marriage Registers	Names of the couple, marriage date, and records of dispensations (if required).	Determined whether heirs could inherit land, titles , or thrones . Enforced prohibitions against consanguinity and affinity
Burial/Death Registers	Deceased person's name, death/burial date .	Fixed the moment when property changed hands or a new reign began.
Bishops' Registers	Appointment of priests, parish visitations, and the granting of marriage licenses and proving of wills.	Documented high-ranking matters like ordinations, benefices, and essential marriage dispensations handled at the diocesan level.
Wills & Inventories	Records of the deceased's last testament , listing possessions and sometimes detailed inventories.	Determined who inherited property (land, titles , thrones). The Church retained jurisdiction over probate until 1858 (in England).

-- The next page is landscape for the size of the table to fit --

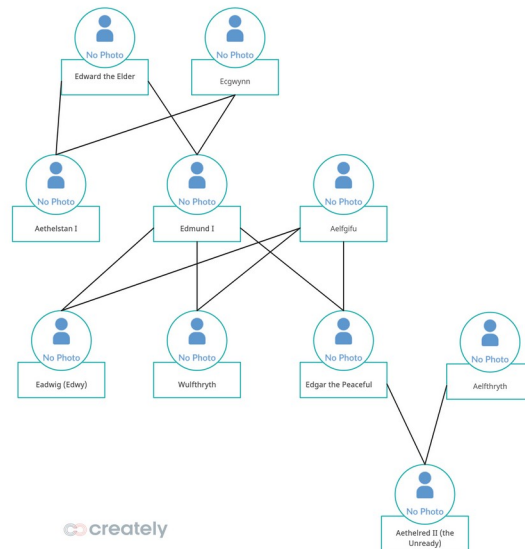
Question 2 (Analysis and Core Database Concepts, 20 points)

Please study the table below which depicts actual data on Aethelstan I, first real king of England (who united the four kingdoms).

name	mother	father	birth_date	death_date	status	bride	groom	marriage_date	marriage_end	end_reason
Edward the Elder	—	—	871-06-01	924-07-17	noble	—	—	—	—	—
Ecgbrynn	—	—	875-03-15	901-08-10	noble	—	—	—	—	—
Aethelstan I	Ecgbrynn	Edward the Elder	894-05-05	939-10-27	noble	—	—	—	—	—
Eadgifu	—	—	902-01-01	966-08-26	noble	—	—	—	—	—
Edmund I	Eadgifu	Edward the Elder	921-01-01	946-05-26	noble	—	—	—	—	—
Aelfgifu (wife of Edmund I)	—	—	922-07-01	944-01-01	noble	—	—	—	—	—
	—	—	—	—	—	Aelfgifu	Edmund I	939-01-01	944-01-01	death (bride)
Eadwig (Edwy)	Aelfgifu	Edmund I	940-01-01	959-10-01	noble	—	—	—	—	—
Edgar the Peaceful	Aelfgifu	Edmund I	943-01-01	975-07-08	noble	—	—	—	—	—
Wulfthryth	—	—	945-01-01	1000-01-01	noble	—	—	—	—	—
Aelfthryth	—	—	945-01-01	1001-11-17	noble	—	—	—	—	—
	—	—	—	—	—	Aelfthryth	Edgar the Peaceful	964-01-01	975-07-08	death (groom)
Aethelred II (the Unready)	Aelfthryth	Edgar the Peaceful	966-01-01	1016-04-23	noble	—	—	—	—	—

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a) (10 points) Draw the genealogy graph (who married who and who was born out of which marriage) based on this table. Do not bother with writing the names exactly correctly.



b) (5 points) If we implemented this table in MySQL what would be the data types for these columns,

name, mother, father, status, bride, groom, end_reason - VARCHAR
 birth date, death date, marriage_date, marriage_end, - DATE

c) (5 points) write the CREATE statement for this table, with proper data types. You do not need to write the INSERT statements for data population. You are required to demonstrate your command of SQL to complement part (b) above.

```

CREATE TABLE
Midterm_Lineage_Unnormalized (
  -- Personal Data Columns
  -- NULL for Marriage Records
  name          VARCHAR(100),
  mother        VARCHAR(100),
  father        VARCHAR(100),
  status        VARCHAR(50),
  -- Date Columns
  -- Using DATE type for efficiency

```

```

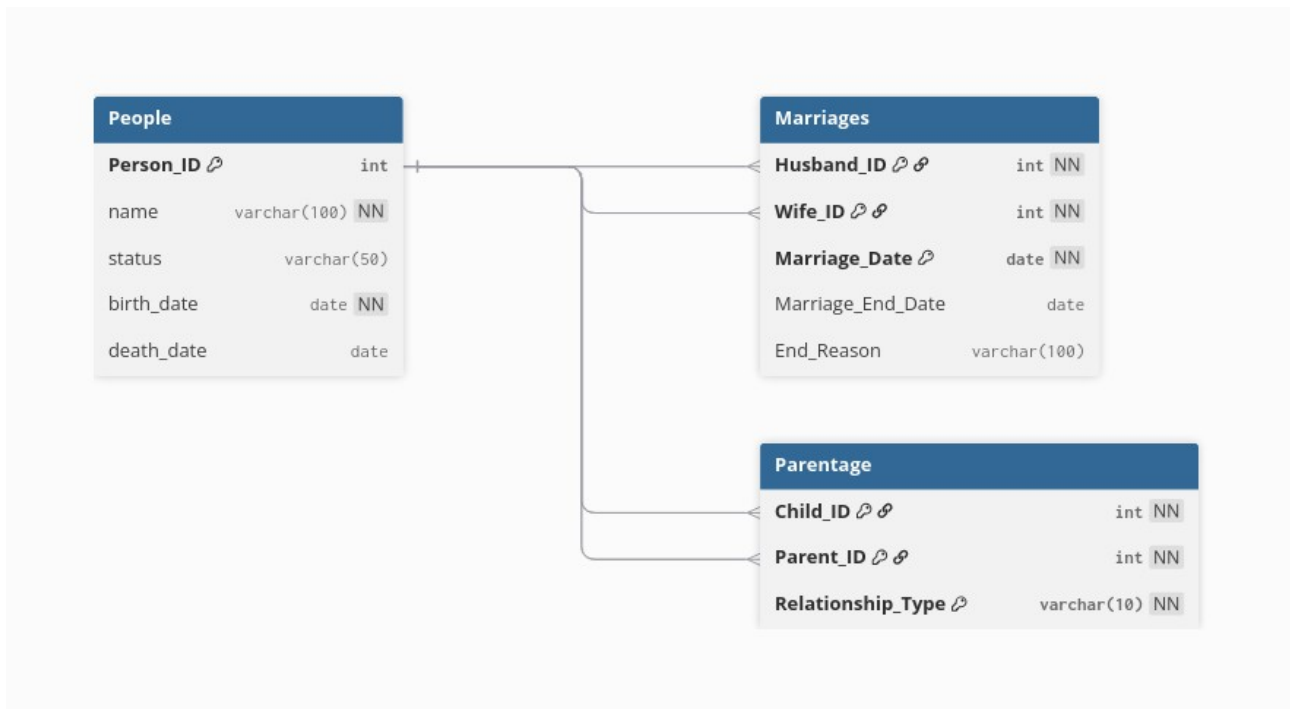
  birth_date    DATE,
  death_date    DATE,
  -- Marriage Data Columns
  -- NULL for Individual Records
  bride        VARCHAR(100),
  groom        VARCHAR(100),
  marriage_date DATE,
  marriage_end  DATE,
  end_reason    VARCHAR(100)
);

```

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d) (10 points) Please normalize this table to 2NF. Add any primary and foreign keys as you'd like. Draw a pseudo ER diagram on the normalized tables. You are not required to create perfect diagram, just enough to understand the structure.

USING BASIC RULES, NORMALIZATION TO 2NF CREATES THREE TABLES: PEOPLE, MARRIAGE AND PARENTAGE. THREE TABLES MEANS THERE IS NO NEED FOR SELF-REFERENCING (RECURSIVE) KEY USE. USING AI TO CREATE DBML and IMPORTING THAT DBML INTO DBDIAGRAM.IO CREATES THE FOLLOWING ER DIAGRAM.

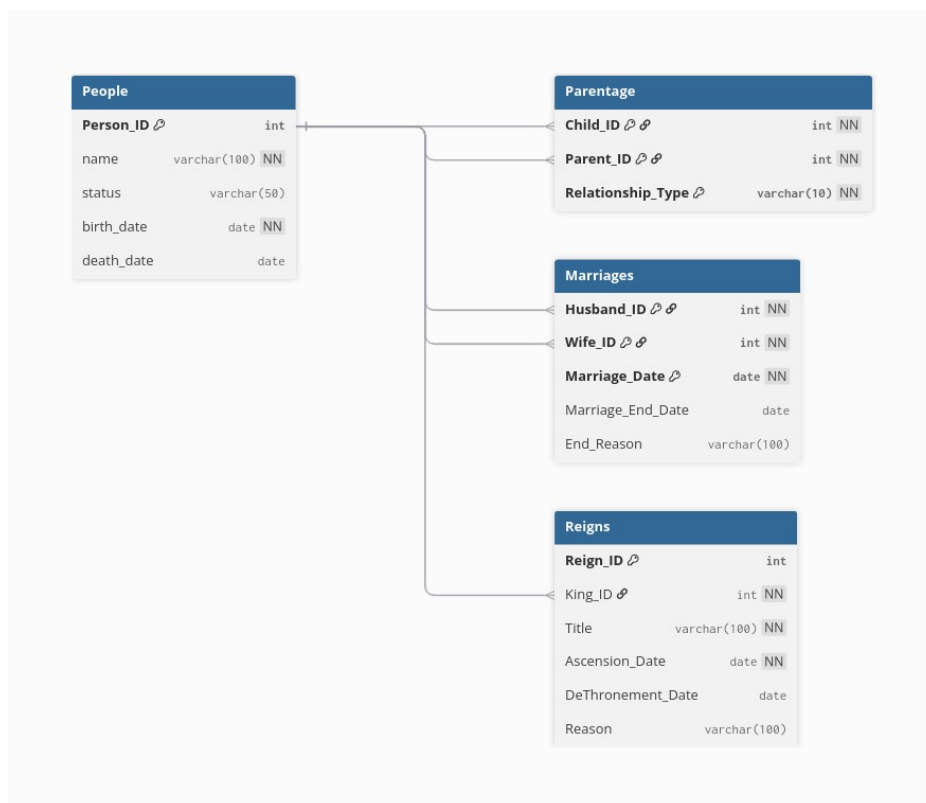


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Question 3 (Analysis and Core Database Concepts, 15 points)

Ascension to throne and being de-throned is common and kings' lineage would be very important records as well. The lineage diagram you created illustrates the ascension with two different options. Aethelstan I died without heirs so his brother ascended. But from that point on there were heirs. The diagram does not illustrate the forceful takeover of the throne, but the English history afterwards is very illustrative of how that happens. In rare cases, more than one king was also present (ie. civil war between those multiple kings).

a) (10 points) Extend these tables by creating a kings table. Assume if someone ascends as queen they will count as king as well. Draw the ER diagram for this table, and shows its relations to other tables. You can summarize the other tables in your diagram. You do not need a perfect diagram, just illustrate your table.



b) (5 points) Write a SQL statement (ie. SELECT) that queries the current king(s) based on the table you created.

```
SELECT
    P.name AS Current_Monarch_Name,
    R.Title AS Reign_Title,
    R.Ascension_Date
FROM
    People P
INNER JOIN
    Reigns R ON P.Person_ID = R.King_ID
```

```
WHERE
    -- The core condition: a current reign is one
    that has no end date defined.
    R.DeThronement_Date IS NULL
ORDER BY
    R.Ascension_Date;
```

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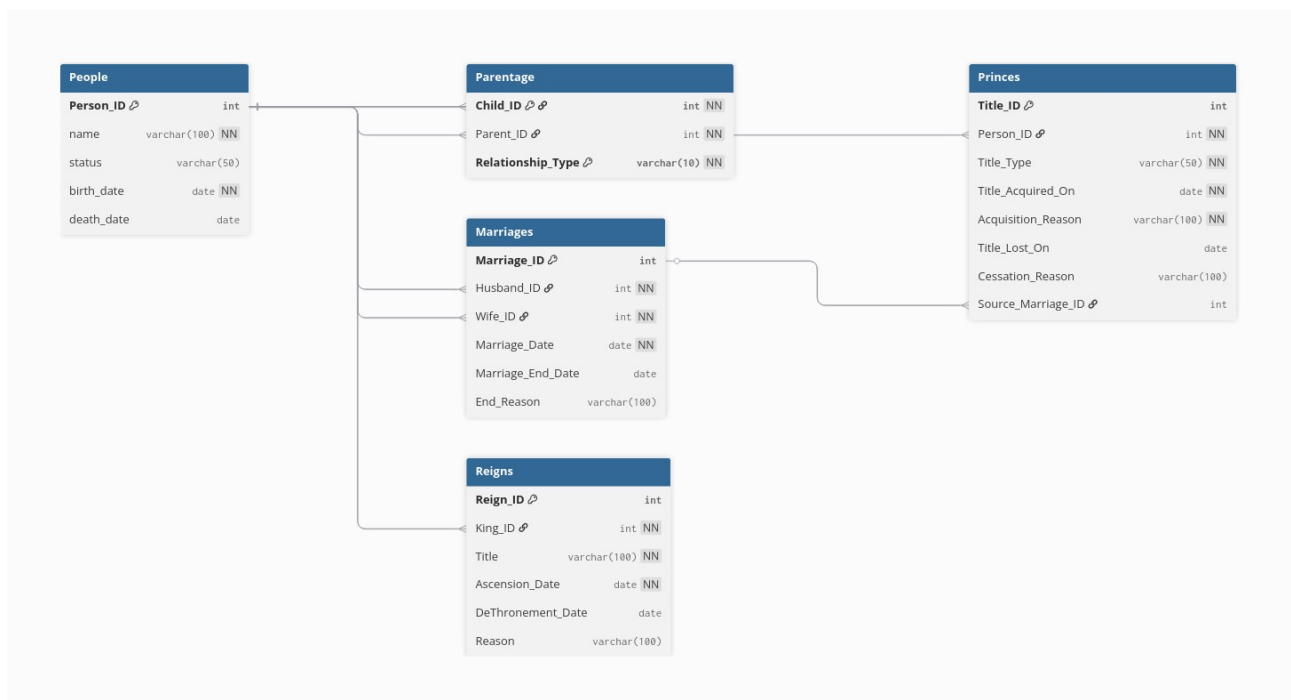
Question 4 (Analysis and Core Database Concepts, 35 points)

Someone could become a prince or princess by birth (as their parent is king at birth) or become one (as their parent becomes a king, or marries into a prince or princess, in rare cases by adoption of king, or also in rare cases by announcement of king).

- If a prince by birth (or adoption) has his parents de-throned, they remain a prince.
- If they become a princess by marriage, and their spouse dies they remain a princess, but they should not re-marry a commoner. If they remarry a commoner their title is removed.
- In contrast princesses who are princess by lineage are always princesses.
- For becoming a prince through marriage, a man who marries a princess who holds the throne in her own right is almost never made prince in the same permanent way a woman is. Their prince status ends with the marriage (divorce, death, etc).
- So there is a distinction between "born" and "made" princes/princesses.

Based on this additional information:

a) (10 points) Extend these tables by creating a princes table to be able to understand who is a prince/princess, since when (at birthdate or at parents' ascension), and why (born or made). Draw the ER diagram for this table, and shows its relations to other tables. You can summarize the other tables in your diagram. You do not need a perfect diagram, just illustrate your table.



b) (5 points) write a SQL statement that creates this table for MySQL.

TWO TABLES WERE SIGNIFICANTLY MODIFIED FROM THE PREVIOUS STRUCTURES: MARRIAGES AND PARENTAGE.

- **THE MARRIAGES TABLE WAS MODIFIED TO ADD A SURROGATE PRIMARY KEY, MARRIAGE_ID.**

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- THE PARENTAGE TABLE'S COMPOSITE PRIMARY KEY WAS MODIFIED TO BE MORE CONCISE AND ACCURATE. ORIGINAL KEY (Q2D): (CHILD_ID, PARENT_ID, RELATIONSHIP_TYPE) MODIFIED KEY (Q4B): (CHILD_ID, RELATIONSHIP_TYPE)

AND THE PRINCES TABLE WAS ADDED:

<pre>CREATE TABLE Princes (Title_ID INT AUTO_INCREMENT PRIMARY KEY, Person_ID INT NOT NULL, Title_Type VARCHAR(50) NOT NULL, Title_Acquired_On DATE NOT NULL, Acquisition_Reason VARCHAR(100) NOT NULL,</pre>	<pre> Title_Lost_On DATE, Cessation_Reason VARCHAR(100), Source_Marriage_ID INT, FOREIGN KEY (Person_ID) REFERENCES People(Person_ID), FOREIGN KEY (Source_Marriage_ID) REFERENCES Marriages(Marriage_ID));</pre>
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PLEASE NOTE: THE USE OF COMPOSITE KEYS IN THIS DESIGN IS ESSENTIAL FOR ENFORCING UNIQUENESS AND MAINTAINING DATA INTEGRITY ACROSS RELATIONSHIP TABLES, AS SURROGATE KEYS ALONE ARE INSUFFICIENT TO DEFINE THE BUSINESS FACTS.

- MARRIAGES TABLE: THE KEY (HUSBAND_ID, WIFE_ID, MARRIAGE_DATE) ENFORCES THAT A COUPLE CAN ONLY BE MARRIED ONCE ON A SPECIFIC DATE. IF ONLY A SURROGATE KEY (MARRIAGE_ID) WERE USED, THE DATABASE WOULDN'T PREVENT THE ACCIDENTAL CREATION OF DUPLICATE MARRIAGE RECORDS FOR THE SAME COUPLE. THE UNIQUE INDEX ON THIS SET OF COLUMNS IS CRUCIAL HERE, EVEN THOUGH A SURROGATE KEY IS PRESENT, BECAUSE IT ENFORCES THE BUSINESS RULE OF UNIQUENESS FOR THE MARRIAGE EVENT.
- PARENTAGE TABLE: THE COMPOSITE KEY (CHILD_ID, RELATIONSHIP_TYPE) IS CRITICAL BECAUSE IT ENFORCES THE RULE THAT EVERY CHILD CAN HAVE ONLY ONE MOTHER AND ONE FATHER. THIS ELIMINATES DATA ENTRY ERRORS WHERE A CHILD MIGHT BE ASSIGNED TWO RECORDS AS 'MOTHER'. THE FOREIGN KEYS ENSURE THAT CHILD_ID AND PARENT_ID LINK TO VALID PEOPLE, WHILE THE PRIMARY KEY GUARANTEES THE NECESSARY UNIQUENESS WITHIN THE LINEAGE STRUCTURE.
- REIGNS TABLE: THE UNIQUE INDEX ON (KING_ID, ASCENSION_DATE) IS NECESSARY BECAUSE A PERSON CAN HAVE MULTIPLE REIGNS (1:N), BUT THEY CANNOT BEGIN TWO DIFFERENT REIGNS ON THE EXACT SAME DATE. THIS PREVENTS OVERLAPPING OR DUPLICATE REIGN RECORDS, PRESERVING THE CHRONOLOGICAL ACCURACY REQUIRED FOR THE HISTORICAL DATA.

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c) (10 points) write a SQL statement that queries the current prince & princesses (ie. alive), their reason (born or married).

<pre>SELECT P.name AS Current_Title_Holder, PR.Title_Type, PR.Acquisition_Reason FROM People P INNER JOIN Princes PR ON P.Person_ID = PR.Person_ID -- Link the person to their titles WHERE -- Condition 1: The person is currently alive P.death_date IS NULL</pre>	<pre>-- Condition 2: The specific title is currently held (has not been lost) AND PR.Title_Lost_On IS NULL -- Condition 3: Filter by the requested acquisition reasons (Born or Married) AND PR.Acquisition_Reason IN ('Born', 'Made (Marriage)') ORDER BY PR.Title_Type, P.name;</pre>
--	--

d) (10 points) write one or more SQL statements that updates the records in case of a "made" princess is re-marrying a commoner and losing princess status.

<pre>UPDATE Princes SET Title_Lost_On = CURRENT_DATE(), -- Sets the cessation date to today (or date of remarriage) Cessation_Reason = 'Remarriage to commoner' WHERE</pre>	<pre>Person_ID = TARGET_PRINCESS_ID -- Identify the princess who remarried AND Title_Type = 'Princess' AND Acquisition_Reason = 'Made (Marriage)' -- Crucial: Only target titles acquired via marriage AND Title_Lost_On IS NULL; -- Only update currently active titles</pre>
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Question 5 (Core Database Concepts, 10 points)

In Question 4, you used more than one table for your queries. Did your SQL statements make use of joins or they employed simple use of WHERE clause? In any case, explain why you used (or not used) joins.

THE SQL STATEMENT FOR QUESTION 4C MAKES USE OF JOINS DUE TO THE DATABASE NORMALIZATION. THE PERSON'S IDENTITY AND LIFE STATUS (NAME, 'DEATH_DATE') ARE STORED EXCLUSIVELY IN THE PEOPLE TABLE, WHILE THE DETAILS OF THEIR NON-SOVEREIGN TITLES ARE STORED IN THE SEPARATE PRINCES TABLE. SINCE THE QUERY NEEDED TO RETRIEVE THE PERSON'S NAME AND CHECK THEIR ALIVE STATUS (FROM 'PEOPLE') SIMULTANEOUSLY WITH CHECKING THE CURRENT STATUS OF THEIR TITLE (FROM PRINCES), THE INNER JOIN ON THE SHARED FOREIGN KEY, P.PERSON_ID = PR.PERSON_ID, WAS MANDATORY. A SIMPLE 'WHERE' CLAUSE ALONE CANNOT ACCESS OR RELATE DATA ACROSS MULTIPLE INDEPENDENT TABLES, MAKING THE JOIN ESSENTIAL FOR LINKING THE BIOGRAPHICAL DATA TO THE TITLE HISTORY ACCURATELY.

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Question 6 (Core Database Concepts, 10 points, Bonus Points)

Assume someone is de-throned (not king anymore) and then by civil war regains the title (is king again). The table you have probably has a single period for being a king.

a) (5 points) Discuss how would you update your kings and princes table(s) to accommodate for this. You do not need to draw ER diagrams, etc. Just discuss verbally. If your table(s) already accommodate this, discuss how it already does.

The current design easily handles a monarch being de-throned and then regaining the title because the Reigns table is structured as a 1:N relationship with the People table (One person can have Many reign records). To update the system, you simply close the first reign by running an UPDATE statement to set the DeThronement_Date on the existing record. Then, you insert a completely new row into the Reigns table with the same King_ID, a new Ascension_Date, and a NULL DeThronement_Date. This separates the two periods of sovereignty into distinct historical records, preserving chronological accuracy.

b) (5 points) Would any previously 1-1 relations become 1-N? And if yes, how would you alter the relation structure? If your table(s) already accommodate this, discuss the existing relations. You do not need to draw ER diagrams, etc. for this part as well.

No, the core relationship does not need to be altered. The relationship between the People and Reigns tables was already established as a 1:N (One person to Many reigns) during the initial design (Q3a). This existing 1:N structure inherently accommodates the business rule of a king regaining a title. No changes are required because the foreign key, Reigns.King_ID, allows the same individual to be linked to multiple, separate reign records, representing their non-contiguous periods of rule.

Question 7 (Core Database Concepts, 10 points, Bonus points)

Discuss shortly if being in 2NF or 3NF makes adding tables, adding columns or altering 1-1 / 1-N status of relations easier or harder.

3NF makes data insertion safer because it eliminates transitive dependencies (where one non-key column depends on another non-key column). This ensures that every fact is stored exactly once, preventing inconsistencies and update anomalies. While 2NF is simpler, it still allows redundancy. The major advantage of 3NF is that when you insert a new record, you only need to provide the primary key and the attribute that directly depends on it, making each insertion atomic and protecting the overall integrity of the historical data structure.